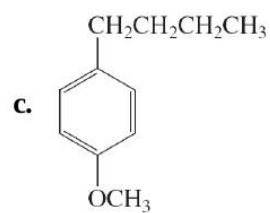
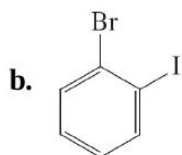
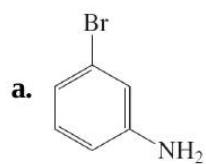


41. Reaction review. Without consulting the Reaction Road Map on [p. 749](#), suggest a combination of reagent and monosubstituted benzene that would give each of the following compounds. (**Hint:** Refer to [Table 16-2](#) for guidance. **Caution:** Directing effects arise from the group already

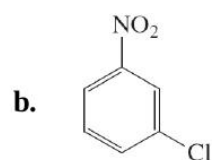
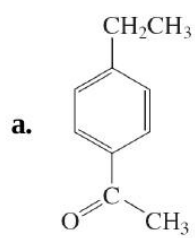
42. Reaction review. The preparation of the following compounds requires more than one step. As in [Problem 41](#), suggest a monosubstituted benzene as starting material and the reagents for all the steps needed to complete their synthesis.



43. Give the expected major product(s) of each of the following reactions.

44. **CHALLENGE** (a) When a mixture containing one mole each of the three dimethylbenzenes (*o*-, *m*-, and *p*-xylene) is treated with one mole of chlorine in the presence of a Lewis acid catalyst, one of the three hydrocarbons is monochlorinated in 100%~~100%~~ yield, whereas the other two remain completely untouched. Which isomer reacts? Explain the differences in reactivity. (b) The same experiment carried out on a mixture of the following three trimethylbenzenes gives a similar outcome. Answer the questions posed in (a) for this mixture of isomers.

45. Propose a reasonable synthesis of each of the following multiply substituted arenes from benzene.



54. Electrophilic substitution on the benzene ring of benzenethiol (thiophenol, $\text{C}_6\text{H}_5\text{SH}$) is not possible. Why? What do you think happens when benzenethiol is allowed to react with an electrophile? (Hint: Review [Section 9-10](#).)

55. **(a)** Although methoxy is a strongly activating (and ortho, para-directing) group, the meta positions in methoxybenzene (anisole) are actually slightly deactivated toward electrophilic substitution relative to benzene. Explain. **(b)** Draw a set of potential-energy curves similar to those in [Section 16-2](#), comparing the electrophilic substitution of benzene with that of methoxybenzene (anisole) in ortho, para and meta positions.

56. Predict the result of mononitration of

